7.5 Structure of DNA

In 1953, J.D. Watson and F.H.C. Crick proposed a precise three dimensional model of DNA structure based on model building studies, base composition and X-ray diffraction studies carried out by Maurice Wilkins and Rosalind Franklin. This model is popularly known as the DNA double helix (Figure7.6).

DO YOU KNOW?

The Nobel prize in Physiology or Medicine(1962) was awarded jointly to Francis Harry Crompton Crick, James Dewey Watson and Maurice Hugh Fredrick Wilkins "for discoveries concerning the molecular structure of nucleic acids



Francis Harry Compton Crick



James Dewey Watson



Maurice Hugh Fredrick Wilkins

7.5.1 Different forms of DNA

There are three forms of DNA – A, B and Z DNA. The characteristics of each form of DNA is listed in the table given below.

Table 7.2 Properties of major foms of DNA

Particulars	A DNA	B DNA	Z DNA
Helix	Right handed	Right handed	Left handed
Base pairs per turn	~11	~10.5	~12
Helical Diameter (nm)	2.6	2.0	1.8
Helical length (nm)	2.6	3.4	3.7
Shape	Broadest	Intermediate	Narrowest
Major Grove	Wide, deep	Narrow, deep	Flat
Minor Grove	Narrow, shallow	Broad, shallow	Narrow, deep

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